

Ex:No: 3

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Comprehensive Statistical Analysis of Retail Sales Data

Problem Description:

A retail organization maintains sales data for 5 products across 12 months. Each product has:

- **Numerical attributes:** MonthlySales, Price
- **Categorical attributes:** ProductName, Category, Region

Due to business operations, the dataset may contain variability in sales and customer regions.

The organization wants to:

1. Analyze **frequency distributions** of products and regions.
2. Calculate **measures of central tendency** and **dispersion** for sales data.
3. Conduct **hypothesis testing** to compare sales across categories and regions.
4. Model data using **statistical distributions** (Binomial, Poisson, Normal, Chi-squared) for probability analysis.

Objective:

To perform **complete statistical analysis** on retail sales data in R by:

- Exploring data structure and identifying trends
- Calculating **mean, median, mode, variance, SD, range**
- Performing **T-Test, ANOVA, F-Test, Mann-Whitney U-Test, Fisher's Exact Test, Kruskal-Wallis Test, Bartlett's Test**
- Applying **Binomial, Poisson, and Normal distributions**

This provides a **framework for business insights and decision-making**.

Procedure:

- **Step 1: Create and Explore Dataset**

```
# Sample retail sales dataset with ProductID, ProductName, Category, Region,  
# MonthlySales and Price
```

```
sales_data <- data.frame(  
  ProductID = 1:8,  
  ProductName = c("Laptop", "Mobile", "Tablet", "Camera", "Headphone",  
    "Mouse", "Keyboard", "Charger"),  
  Category = c("Electronics", "Electronics", "Electronics", "Electronics",  
    "Accessories", "Accessories", "Accessories", "Accessories"),  
  Region = c("North", "South", "East", "West", "North", "South", "East",  
    "West"),  
  MonthlySales = c(50, 120, 80, 30, 150, 70, 60, 55),  
  Price = c(700, 300, 250, 400, 50, 20, 30, 25)  
)
```

str(sales_data)

'data.frame': 8 obs. of 6 variables:

dfProductid : int 1 2 3 4 5 6 7 8

dfProductName : chr "laptop" "mobile" "tablet" "canon" ,...

dfCategory : chr "electronics" "electronics" "electronics" "electronics" "Accessories" ,...

dfRegion : chr "North" "South" "East" "West" "North" "South" "East" "West" ,...

dfMonthlySales : num 50 150 80 90 150 70 60 55

dfPrice : num 1100 900 750 400 50 20 20 30 25

> table(sales_data\$Category)

Accessories 1
Electronics 4

> table(sales_data\$Region)

East 2
North 2
South 2
West 2

> print(mean_sales)

[1] 76.875

> print(median_sales)

[1] 65

> print(mode_sales)

[1] 30 50 55 60 70 80 100 150

Summary:

Product ID	Product Name	Category	Region	Monthly Sales
1	laptop	electronics	North	50
2	mobile	electronics	South	150
3	tablet	electronics	East	80
4	canon	electronics	West	90
5	accessories	Accessories	North	150
6	accessories	Accessories	South	70
7	accessories	Accessories	East	60
8	accessories	Accessories	West	55

Explore data structure sales_data. Use str()

str(sales_data)

Summary statistics sales_data

summary(sales_data)

Frequency distributions (Category and Region. Use table())

table(sales_data\$Category)

table(sales_data\$Region)

• Step 2: Measures of Central Tendency

Mean, Median, Mode for Monthly Sales

mean_sales = mean(sales_data\$MonthlySales)

median_sales = median(sales_data\$MonthlySales)

mode_sales = mfv(sales_data\$MonthlySales)

Print results mean_sales, median_sales, mode_sales

print(mean_sales)

print(median_sales)

print(mode_sales)

> print (nummber_sales)

[1] 1568.839

> print (cd_sales)

[1] 39.54541

> print (range_sales)

[1] 30 150

> print (numerical_data)

Product Id	MonthlySales	Price
1	50	700
2	120	300
3	80	250
4	30	400
5	150	50
6	70	80
7	60	30
8	55	85

> cor (sales_data \$ MonthlySales, sales_data \$ Price, method = "pearson")
[1] -0.3179059.

• Step 3: Measures of Dispersion

Variance, Standard Deviation and Range of sales for MonthlySales

variance_sales = var (sales_data \$ MonthlySales)

sd_sales = sd (sales_data \$ MonthlySales)

range_sales = range (sales_data \$ MonthlySales)

Print results variance_sales; sd_sales; range_sales

print (variance_sales)

print (sd_sales)

print (range_sales)

• Step 4: Correlation Analysis for Retail Sales Data

◦ Correlation Coefficient: To analyze the relationship between Monthly Sales and Product Price using correlation analysis

Select numerical variables

numerical_data <- sales_data[, c("Product Id", "MonthlySales", "Price")]

print (numerical_data)

Compute Pearson correlation

correlation_coef = cor (sales_data \$ MonthlySales, sales_data \$ Price, method = "pearson")

Print correlation_coef

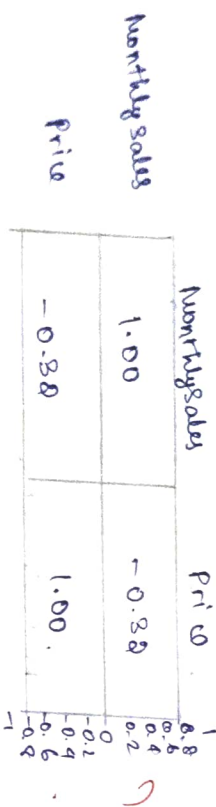
print (correlation_coef)

- A positive value indicates higher-priced products tend to have higher sales
- A negative value indicates higher prices reduce sales
- A value near zero indicates weak or no linear relationship

> priu (cor-matrix).

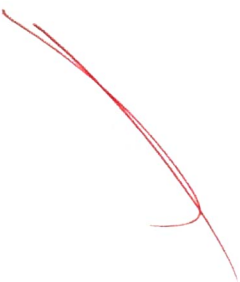
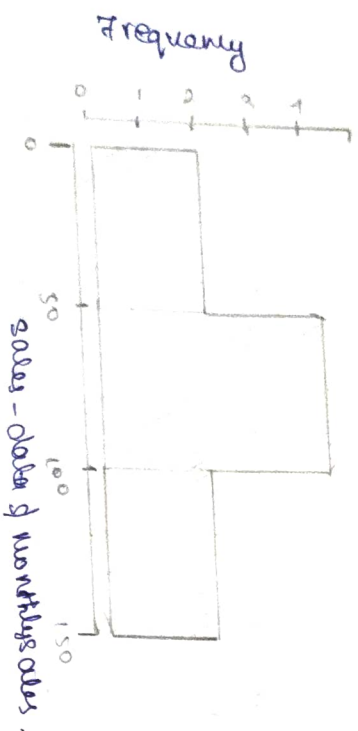
MonthlySales Price
1.0000000 -0.3179059
Price -0.3179059 1.0000000

> corplot (cor (numerical-data[2:8]), method = 'number').



> hist (sales-data & monthlysales).

Histogram of sales-data & monthlysales



- Correlation Matrix for Retail Sales Data: To analyze pairwise relationships among numerical variables using a correlation matrix
- # Compute correlation matrix
- cor-matrix <- cor (sales-data[3:8], sales-data, is.numeric=T)

Display correlation matrix

priu (cor-matrix).

- Visualizing Correlation Matrix:
- # Install and apply corplot for correlation matrix

install.packages('corplot')

library (corplot)

corplot (cor (numerical-data[2:8]), method = 'number')

- Step 5: Data Visualization
- Histogram of Monthly Sales: To understand the distribution and frequency pattern of monthly sales

hist (sales-data & monthlysales).

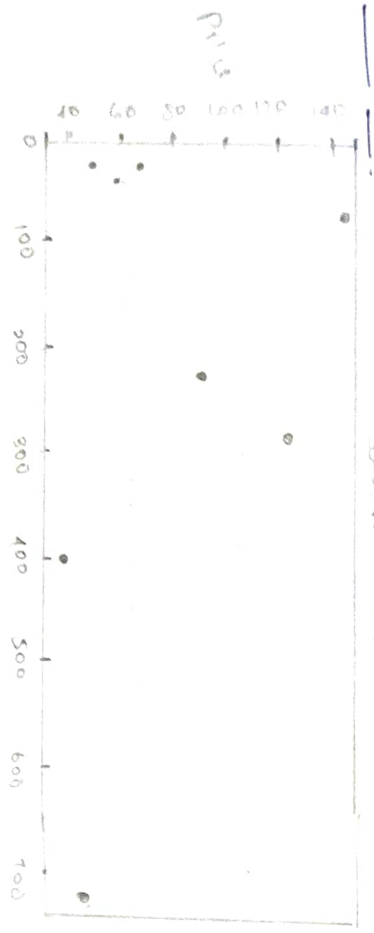


boxplot

sales_data & MonthlySales



Scatter plot:



T-test:

data: sales - accessories and sales - electronics.

$t = 0.46332$, $df = 5.901$, $p\text{-value} = 0.6597$.

alternative hypothesis: true difference in means is not equal to 0.

95 percent confidence interval:

-89.16347 86.66347

sample estimates:

mean of x mean of y
83.75 70.00

Boxplot of Monthly Sales by Category: To compare sales distribution between Electronics and Accessories and identify outliers

box plot (sales_data & MonthlySales ~ sales_data\$category)

Scatter Plot for Correlation Visualization: To visually confirm the correlation between price and sales

Plot (sales_data & MonthlySales ~ sales_data\$Price, pch=20, main="Sales vs Price", xlab="sales", ylab="Price")

Step 6: Hypothesis Testing

T-Test: Compare Accessories vs Electronics:

Category = "Accessories", sales_data & MonthlySales [sales_data\$Category == "Accessories"]

Category = "Electronics", sales_data & MonthlySales [sales_data\$Category == "Electronics"]

t.test (sales_data\$Accessories, sales_data\$Electronics)

ANOVA:

call: $\text{aov}(\text{formula: sales-data} \sim \text{monthlysales} \sim \text{sales-data} \& \text{region})$

terms:

	sales-data\$region	residuals
sum of squares	4184.875	6660.500
df	3	4

residual standard error: 41.11721

F-statistics may be unbalanced.

F-test:

data: sales-accessories and sales-electronics.

$F = 1.9976$, num df = 3, denom df = 3, P-value = 0.8386

alternative hypothesis: true ratio of variances is not equal to 1

95 percent confidence interval:

0.08404294 20.08317299

sample estimates:

ratio of variances

1.997554

Mann-Whitney U-test:

data: sales-accessories and sales-electronics.

$W = 10$, P-value = 0.6857

alternative hypothesis: true location shift is not equal to 0

Fisher's test:

data: contingency-table

P-value = 1

alternative hypothesis: two.sided.

Kruskal-Wallis rank sum test:

data: sales-data\$monthlysales by sales-data\$region

Kruskal-Wallis chi-squared = 3, df = 3, P-value = 0.3946

ANOVA: Compare sales across regions:

$\text{aov}(\text{sales-data} \sim \text{monthlysales} \sim \text{sales-data} \& \text{region})$

F-Test: Compare Accessories vs Electronics:

var.test(sales-data\$^{Accessories}, sales-Electronics).

Mann-Whitney U-Test: Compare Accessories vs Electronics:

wilcox.test(sales-accessories, sales-electronics).

Fisher's Exact Test: Category and Region

contingency-table <- table(sales-data\$categories, sales-data\$region)

fisher.test(contingency-table).

Kruskal-Wallis Test: Compare sales across regions:

kruskal.test(sales-data\$monthlysales ~ sales-data\$region).

Bartlett's Test: Compare sales across regions:

bartlett.test(sales-data\$monthlysales ~ sales-data\$region).

Bartlett test:

data : sales - data & monthly sales by sales - data & region.

Bartlett's χ^2 -squared : 8.1460, df = 3, p-value = 0.5986.

binomial:

[1] 0.8373046875 0.395507812 0.263671875 0.857890625

0.0146484375 0.009765625

Poisson:

[1] 0.000458823 0.003524896 0.013548819 0.87418848

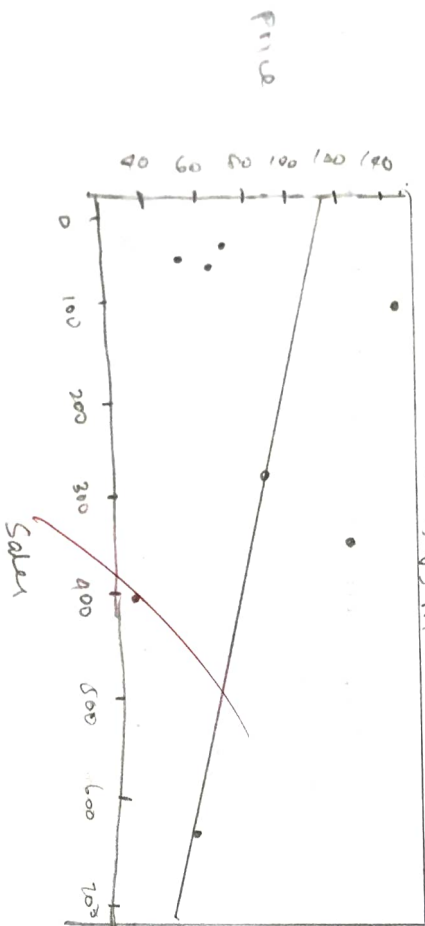
0.06672586 0.102590127.

normal - distribution:

[1] 0.008607992 0.005866418 0.0100678 0.004097033

0.0018282933 0.009986900 0.009216280 6.001657064.

> abline(lam (sales - data & monthly sales ~ sales - data & price), lty = "lod", lwd = 2)



• Step 5: Statistical Distributions

◦ Binomial Distribution

- Scenario: Probability of having a "high sales" month (success = MonthlySales > 100) in 5 months

Identify high sales months

high-sales <- ifelse (sales - data & monthly sales > 100, 1, 0)

Binomial probability for 0 to 5 high-sales months

dbinom(0:5, 5, 20/25, Prob = mean (high-sales)).

◦ Poisson Distribution

- Scenario: Model the probability of a number of sales events per month (λ = average MonthlySales / 10)

lambda <- mean (sales - data & monthly sales) / 10

dpois (0:5, lambda = lambda).

◦ Normal Distribution

- Scenario: Model MonthlySales as normally distributed. find probability density for each product.

mean-sales <- mean (sales - data & monthly sales)

sd-sales <- sd (sales - data & monthly sales)

dnorm (sales - data & monthly sales, mean = mean sales

sd = sd-sales)

Result:

Thus, the program successfully demonstrates:

- Frequency distributions for categorical attributes (Category, Region)
- Measures of central tendency and dispersion for numerical attributes (MonthlySales)
- Performing hypothesis testing (T-Test, ANOVA, F-Test, Mann-Whitney U-Test, Fisher's Exact Test, Kruskal-Wallis Test, Bartlett's Test)
- Modeling data using Binomial, Poisson, and Normal distributions

This provides a complete statistical framework for analyzing retail sales data and supports data-driven business decisions.